Certified Personal Trainer

DOMAIN TWO

NUTRITION

EXAM WEIGHT 10%

ISSA INTERNATIONAL SPORTS SCIENCES ASSOCIATION

It is outside the scope of practice for a Certified Personal Trainer to prescribe meal plans or supplements to clients. However, trainers can review a client's eating habits and make suggestions regarding ways they can improve and eat to support their fitness goals.

Although trainers should have a strong understanding of nutrition when working with clients, each state has specific regulations that dictate the legality of offering recommendations on nutrition and supplementation. Any time a trainer is working with a client who has nutrition deficiencies or a chronic condition with specific dietary needs, they should always refer to a dietitian and work with the physician.

Macronutrients - Chapter 14

- Carbohydrates
  - The primary source of energy for the body, in most cases
  - Required in large quantities in the diet to support function and energy production
  - 4 calories per gram
  - Glucose: a simple sugar the body uses for energy production on the cellular level
  - Glycogen: the stored form of glucose found in the liver and muscles
  - Intake recommendation for adults: 45 to 65 percent of total daily calories
  - There are two types of carbohydrates: simple and complex
    - Simple
      - Monosaccharides: any of the class of sugars that cannot be hydrolyzed to give a simple sugar
      - Disaccharides: any of a class of sugars with molecules that contain two monosaccharide residues
    - Complex
      - Starches and fibers
      - Whole-grain breads and cereals
      - Starchy vegetables
      - Legumes
  - Glycemic index (GI): a system that ranks foods on a scale from 1 to 100 based on their effect on blood sugar levels
    - Low GI: 1 to 55
    - Medium GI: 56 to 69
    - High GI: 70 and above

- Fiber: a type of carbohydrate derived from plant-based foods that the body is unable to break down
  - Types of fiber: soluble and insoluble
    - Soluble
      - Dissolves in water and absorbs water from partially digested food
      - Helps slow the digestion process and regulate blood glucose levels, which leads to lower levels of low-density lipoprotein (LDL) cholesterol
    - Insoluble
      - Does not dissolve in water and, instead, adds bulk to the stool
      - Helps food move through the digestive tract
  - Intake recommendations for adults
    - Women: 25 grams
    - Men: 38 grams

- Fats (Lipids)
  - Necessary for vital bodily functions
  - 9 calories per gram
  - Fatty acids: the smaller, absorbable building blocks of the fat that is found in the body
  - Triglyceride: a chemical compound formed when three fatty acids combine with glycerol; the most abundant fat in the body
  - Intake recommendations for adults: 30–35 percent of daily calorie intake
  - Saturated fat
    - Solid at room temperature
    - Considered to be a detrimental fat; can cause cholesterol buildup in the arteries and increase the risk for heart disease
    - Food sources include butter, solid shortening, lard, cheese, red meat, ice cream, coconut oil, and palm oil
  - Unsaturated fat
    - Liquid at room temperature
    - Types
      - Monounsaturated
      - Polyunsaturated
      - Trans fat
        - Detrimental fat
        - Elevates LDL and lowers HDL

- Essential fatty acids
  - Omega-6 fatty acids: a family of pro-inflammatory and anti-inflammatory polyunsaturated fatty acids that have in common a final carbon-carbon double bond
  - Omega-3 fatty acids: an unsaturated fatty acid occurring chiefly in fish oils
  - Eicosapentaenoic acid (EPA): a fatty acid found in fish and fish oils, which is believed to lower cholesterol, especially cholesterol bound to low-density lipoproteins
  - Docosahexaenoic acid (DHA): an omega-3 fatty acid that is a primary structural component of the human brain, cerebral cortex, skin, sperm, testicles, and retina

- Protein
  - Necessary for building and maintaining body tissues
  - 4 calories per gram
  - Amino acid: building blocks of protein

9 ESSENTIAL AMINO ACIDS	NON-ESSENTIAL AMINO ACIDS
Cannot be made by the body, must be obtained from the diet.	Not essential in the diet as the body can synthesize it.
Isoleucine (BCAA)	Alanine
Leucine (BCAA)	Arginine
Histidine	Asparagine
Lysine	Aspartic acid
Methionine	Cysteine
Phenylalanine	Glutamic acid
Threonine	Glutamine
Tryptophan	Glycine
Valine (BCAA)	Proline
	Serine
	Tyrosine

  - Incomplete and complete proteins
    - Complete protein: a food source that contains each of the nine essential amino acids
      - Example: Eggs
    - Incomplete proteins: a food source that lacks one or more of the nine essential amino acids
      - Example: Peas

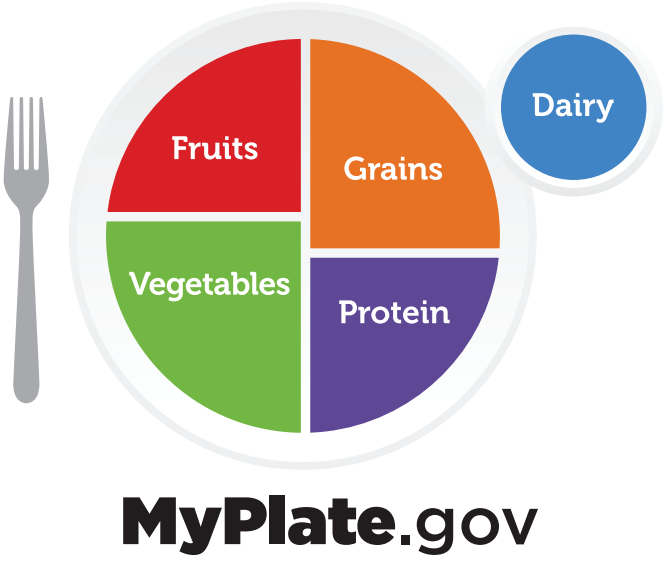
- Water
  - Water makes up around 60 percent of total body weight in adults and about 75 percent in children.
  - Intake recommendations: between 91 and 125 fluid ounces or 2.7 to 3.7 liters of water per day is recommended for adults
    - About 20 percent of that intake comes from food.
  - The human body relies on water to function properly.
  - 3-3-3 Rule
    - The body can go
      - 3 minutes without air,
      - 3 days without water,
      - and 3 weeks without food.
  - Key terms
    - Electrolytes: minerals in the body that have an electric charge.
    - Intracellular fluid (ICF): water found within the cells of the body.
      - Approximately two-thirds of the body's total water is found within the cell
    - Extracellular fluid (ECF): water found outside the cells and between tissues.
      - Approximately one-third of the body's total water is found outside the cells
    - Dehydration: a harmful loss or removal of water in the body.

Micronutrients - Chapter 14

- Micronutrients: substances required in small quantities in the diet for optimal body functioning (vitamins, minerals, antioxidants)
  - Some minerals are required in large quantities (macrominerals), while others are only required in trace amounts (trace minerals)
  - Vitamins
    - Fat-soluble
    - Water-soluble
  - Most common vitamin deficiencies in the US
    - Vitamin B7
    - Vitamin D
    - Vitamin E
    - Chromium
    - Iodine
    - Molybdenum

Dietary Guidelines and My Plate - Chapter 14

- Dietary Guidelines for Americans: guidelines for healthy, lifelong eating habits for Americans two years of age and older
  - The current dietary guidelines are useful for making healthy food choices and crafting a well-rounded eating pattern
    - Concept one: following a healthy eating pattern across the lifespan
    - Concept two: focusing on variety, nutrient density, and food amounts
    - Concept three: limiting calories from added sugars and saturated fats and reducing sodium intake; a healthy eating pattern should be low in added sugars, saturated fats, and sodium
    - Concept four: shifting to healthier food and beverage choices
    - Concept five: supporting healthy eating patterns for all
  - Nutrition coaches and fitness professionals can use these guidelines as the basis for helping clients develop healthier eating habits to support their goals
- MyPlate: the current visual nutrition guide published by the USDA Center for Nutrition Policy and Promotion; replaced the food pyramid (outdated)
  - 50% of plate should be fruits and veggies
  - 25% of plate should be whole grains
  - 25% of plate should be lean protein
  - Dairy products should be fat-free or low-fat



Food Labels and Portion Size - Chapter 14

Teaching clients how to understand what and how much they are consuming is an important component of success. Learning how to read food labels and measuring proper portion sizes is one of the best places to start.

- Nutrition facts: a label required by the Food and Drug Administration (FDA) on most food and beverages that details the food's nutrient content
- Ingredient list: a list provided on a food label of each ingredient in a product in descending order of prominence
- Daily value (DV): reference amounts expressed in grams, milligrams, or micrograms of nutrients to consume or not to exceed each day
- The footnote of the label reads "*The % daily value indicates how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice."
    - The percent daily value helps consumers understand the nutrition information in the context of a total daily diet of 2,000 Calories, unless otherwise specified on the label.
- Serving size
    - Package size affects what people eat, so for packages that are between one and two servings—such as a 20-ounce soda or a 15-ounce can of soup—the calories and other nutrients will be required to be labeled as one serving
    - If the package is larger, the addition of another column with the words "per package" details the nutritional values if the entire package is consumed at once, as well as the reference amount for a single serving
- Portion size
    - Personal trainers can help clients easily establish how much food should be consumed using their own hand as the measuring device
        - Protein portions should be about the size of the palm of the hand
        - Vegetable portions should be about the size of a fist
        - Carbohydrate portions should be about the size of a cupped hand
        - Fat portions should be about the size of a thumb

MALES—FOR EACH MEAL
2 palms of protein-dense food
2 fists of vegetables
2 cupped handfuls of carbohydrates
2 thumbs of fat-dense foods
FEMALES—FOR EACH MEAL
1 palm of protein-dense food
1 fist of vegetables
1 cupped handful of carbohydrates
1 thumb of fat-dense foods

Current Diet Trends - Chapter 14

Some diets have delivered everything they promise. The World Health Organization (WHO) has even recognized some popular diets as sustainable and healthy lifestyle choices. However, others fall by the wayside because they don't deliver results or are harmful. To help guide clients, a fitness professional must be aware of popular trending diets and their risks and benefits.

- Detox diet
  - Short term, this can be beneficial; but detoxing for extended periods is contraindicated
  - Examples: colon cleanse, juice cleanse, and liver detox
- Low and no-fat diets
  - Can initially lead to weight loss if calorie guidelines are followed
  - May decrease plasma cholesterol levels and health issues for special populations like pregnant or lactating women, children, and the elderly
  - Dietary fat is important for hormone production and regulation
    - A low-fat diet can upset hormone balance and cause health problems in both men and women
- Blood type diet
  - Blood type A: consume fruit, vegetables, tofu, seafood, soy, and whole grains and avoid most meats; ideally, the diet is mostly vegetarian, with some fish, and largely organic and fresh because of a sensitive immune system
  - Blood type B: choose green, non-starchy vegetables (spinach, asparagus, green beans, artichokes), lean

meats (except chicken), and low-fat dairy while avoiding corn, wheat, lentils, tomatoes, and peanuts; because of digestive sensitivity, this diet is supposed to be largely gluten-free
    - Blood type AB: focus on tofu, seafood, dairy, and green leafy vegetables while avoiding caffeine, alcohol, and cured meat; these individuals supposedly have low stomach acid
    - Blood type O: eat a diet of lean meats, poultry, fish, and vegetables while avoiding grains, beans, and dairy
- Raw-food diet
    - The main claim of proponents is that the diet improves enzyme activity and digestion, but this remains unproven
    - Critics have a lot of concerns, including the risk of eating raw and undercooked meat and animal products
- Low-energy diet (LED) and very low-energy diet (VLED)
    - Physician-supervised
    - A low-calorie diet is considered 800–1,200 calories daily
    - A very low-calorie diet is less than 800 calories daily
- Plant-based diets

PLANT- BASED TYPE	MEAT	GAME	POULTRY	FISH	EGGS	DAIRY	NOTHING ANIMAL-DERIVED
Vegan	✗	✗	✗	✗	✗	✗	✓
Lacto-vegetarian	✗	✗	✗	✗	✗	✓	✗
Ovo-vegetarian	✗	✗	✗	✗	✓	✗	✗
Lacto-ovo-vegetarian	✗	✗	✗	✗	✓	✓	✗
Pescatarian	✗	✗	✗	✓	✓	✓	✗
Pollotarian	✗	✗	✓	✗	✓	✓	✗
Flexitarian	✗	✗	Some	Some	✗	✗	✗

- Keto Diet
    - 50 grams of carbohydrates based on a 2,000-calorie diet
    - Leads to a state of ketosis: a metabolic process that occurs when the body does not have enough carbohydrates for energy; the liver metabolizes fatty acids to produce ketones as a replacement energy source
    - Can cause headache, fatigue, bad breath, constipation, and dehydration
- Atkins diet
    - Similar to the keto diet but carbs are slowly reintroduced
- Carnivore diet
    - Beef, lamb, pork, veal, chicken, and eggs are staples
    - Avoid vegetables, low-lactose dairy, fruits, legumes, nuts, seeds, and grains

- Paleo diet
  - Sometimes referred to as “clean eating”
  - Differs from a carnivore diet by including fruits, vegetables, nuts, and seeds
- Gluten-free diet
  - In individuals with celiac disease, gluten triggers an immune response that damages the lining of the small intestines
  - Avoid anything with wheat, barley, or rye
  - Avoiding gluten isn’t necessary for anyone without a gluten allergy or celiac disease
- DASH diet
  - DASH = Dietary Approaches to Stop Hypertension
  - Can be ideal for someone with high blood pressure with a goal of reducing hypertension with fewer or no medications
  - There are no special food restrictions, just a weekly plan of a healthy 2,000 calories per day
  - Focuses on low-fat dairy, fish, poultry, beans, nuts, and vegetable oils while avoiding fatty meats, full-fat dairy, coconut and palm oil, and sweets
- Mediterranean diet
  - Widely recognized and accepted by the WHO
  - Moderate portions of fruits, vegetables, whole grains, legumes, potatoes, nuts, seeds, olive oil, fish, poultry, dairy, and eggs, with minimal red meats
- Intermittent fasting
  - Focuses on timing of food intake as opposed to the types of food consumed
    - 5/2 - reducing calorie intake for 2 days each week
    - 16/8 - fasting for 16 hours out of a 24 hour day

Supplementation - Chapter 15

A dietary supplement is a product containing one or more dietary ingredients that is intended to supplement a person’s nutrition plan.

TYPE	NUTRITIONAL SUPPLEMENTS	PERFORMANCE SUPPLEMENTS	ERGOGENIC AIDS	BOTANICAL SUPPLEMENTS
Description	Natural or synthetic nutrients intended to supplement a nutrition plan	Vitamins, minerals, proteins, herbs, and amino acids consumed to promote metabolic and physical performance and recovery	Compounds consumed to specifically enhance performance during exercise or competition	Made from plant parts or extracts and intended to supplement a nutrition plan

- Ergogenic aids (performance enhancers)
  - Some ergogenic aids have been banned for use by many organized sports organizations to prevent unfair advantages
  - Anabolic-androgenic steroids (AAS): synthetic variations of the male sex hormone testosterone
  - Creatine: the goal of supplementing with CM is to increase intramuscular levels of creatine and speed the regeneration of creatine phosphate (CP), which is expended during energy production
  - Beta-alanine (BA): a nonessential amino acid that is naturally produced by the body
    - May lower exercise-induced acidosis, potentially delaying fatigue and improving performance
  - Branched-chain amino acids (BCAAs): a group of three essential amino acids—leucine, isoleucine, and valine—that help the body build muscle and decrease muscle fatigue
    - Glutamine
      - Most abundant amino acid in the body
  - Caffeine: reduces perceived pain and exertion and increases fat metabolism
  - Stimulants: a class of drugs that temporarily improve physical or mental function
    - Can speed up body systems and elevate alertness, mood, and awareness
    - Can raise heart rate, temperature, blood sugar, and blood pressure
- Botanical Supplements
  - A botanical is a plant or plant part containing various medicinal properties

BOTANICAL	PRIMARY USAGE	RESEARCH	NOTES
Cannabidiol (CBD)	Reduce anxiety, improve sleep, and increase weight loss	Most research is done on mice, not humans. Results are inconclusive.	Dosages vary depending on form and use.
Cinnamon	Lower blood sugar by increasing insulin sensitivity	Proven effective as an antidiabetic compound and may reduce cholesterol	Taking 2 g/day shows anti-inflammatory properties and lower blood sugar levels.
Echinacea	Build up the immune system and protect against allergies	Results vary but show a reduction in length of sickness with supplementation.	No optimal dose or type
Garlic	Antioxidant and anti-inflammatory properties that protect against allergies and build immunity	Consistent findings for increasing HDL and decreasing LDL by 10–15 percent reduces blood pressure and frequency of the common cold.	Can be supplemented via a nutrition plan or aged garlic capsules. Taking 300 mg/day has been shown to manage hypertension.

BOTANICAL	PRIMARY USAGE	RESEARCH	NOTES
Ginger	Improve the movement of food in the stomach for gut health and increase testosterone	No research regarding gut health, increase in testosterone found in infertile men	May increase HDL and decrease LDL. Reduces inflammatory markers. Comes in forms such as fresh or dried roots, tablets, and liquid extract.
Green tea	Increase norepinephrine and induce fat loss	Fat loss effect is minor and unreliable	May reduce LDL, lower fasting blood glucose, increase oxygen uptake, and reduce muscles soreness
Hoodia	Increase fat loss	Does not suppress appetite, may increase blood pressure and heart rate	Has a desired effect on appetite and weight loss. Little available research in humans.
Milk thistle	Promote cell repair for liver health and detoxification	No research to support claims	May reduce acne lesions with direct application to the skin
Saw palmetto	Increase testosterone levels and improve prostate health	Not effective at increasing testosterone or suppressing prostate growth	Mixed and inconsistent findings regarding sexual function
Valerian	Improve sleep and reduce anxiety	Research points to no effect.	May reduce menstrual pain and symptoms of PMS. Commonly used before bedtime.
Note: g = gram; HDL = high-density lipoprotein; LDL = low-density lipoprotein; mg = milligram *reference the vitamin and mineral tables earlier in this chapter*			

Important nutritional intake terms:

- Estimated average requirement (EAR): the average daily nutrient intake level that is estimated to meet the requirement of half the healthy individuals in a specific life stage or gender
- Dietary reference intakes (DRIs): a set of standards estimating how much of a nutrient should be ingested; used for planning eating patterns for healthy individuals

- Recommended daily allowance (RDA): the average amount of a nutrient adequate to meet the nutrient needs of healthy individuals each day
- Fortified: having had vitamins or other supplements added so as to increase the nutritional value.

RDA (VITAMINS)

VITAMIN	PURPOSE	FOODS		RDA
Biotin	• Energy storage • Protein, carbohydrate, and fat metabolism	Avocados Cauliflower Eggs Fruits Whole grains	Liver Pork Salmon	30 mcg
Folate/Folic acid *Important for pregnancy*	• Prevents birth defects • Protein metabolism • Red blood cell formation	Asparagus Avocado Beans and peas Enriched grain products	Green leafy vegetables Orange juice	400 mcg DFE
Niacin	• Cholesterol production • Converts food into energy • Digestion • Nervous system function	Beans Beef Enriched grain products Nuts	Pork Poultry Seafood Whole grains	16 mcg
Pantothenic acid	• Converts food into energy • Fat metabolism • Hormone production • Nervous system function • Red blood cell formation	Avocados Beans and peas Broccoli Eggs Milk Mushrooms	Poultry Seafood Sweet potatoes Whole grains Yogurt	5 mcg
Riboflavin	• Converts food into energy • Growth and development • Red blood cell formation	Eggs Enriched grains Meats Milk	Mushrooms Poultry Seafood Spinach	1.3 mcg
Thiamin	• Converts glucose into energy • Nervous system function	Beans and peas Enriched grain products Nuts	Pork Sunflower seeds Whole grains	1.2 mcg

VITAMIN	PURPOSE	FOODS		RDA
Vitamin A	• Growth and development • Immune function • Reproduction • Red blood cell formation • Skin and bone formation • Vision	Cantaloupe Carrots Dairy products Eggs Fortified cereals	Green leafy vegetables Pumpkin Red peppers Sweet potatoes	900 mcg RAE *Current RDAs from the DRI reports for vitamin A are expressed as micrograms of retinol activity equivalents (RAE).*
Vitamin B_6	• Immune function • Nervous system function • Protein, carbohydrate, and fat metabolism • Red blood cell formation	Chickpeas Fruits (other than citrus) Potatoes	Salmon Tuna	1.7 mg
Vitamin B_{12}	• Converts food into energy • Nervous system function • Red blood cell formation	Dairy products Eggs Fortified cereals	Meats Poultry Seafood	2.4 mcg
Vitamin C	• Antioxidant • Collagen and connective tissue formation • Immune function • Wound healing	Broccoli Brussels sprouts Cantaloupe Citrus fruits and juices	Kiwi Peppers Strawberries Tomatoes and tomato juice	90 mg
Vitamin D **Deficient in most Americans**	• Blood pressure regulation • Bone growth • Calcium balance • Hormone production • Immune function • Nervous system function	Eggs Fish Fish liver oil Fortified cereals Fortified dairy	Fortified margarine Fortified orange juice Fortified soy beverages Sunlight	20 mcg

VITAMIN	PURPOSE	FOODS		RDA
Vitamin E	• Antioxidant • Formation of blood vessels • Immune function	Fortified cereals Fortified juices Green vegetables Nuts and seeds	Peanuts Vegetable oils Wheat germ	15 mg AT *Current RDAs from the DRI reports for vitamin E are expressed as milligrams of alpha-tocopherol (mg AT).*
Vitamin K	• Blood clotting • Strong bones	Broccoli Collard greens Kale	Spinach Turnip greens	120 mcg
Note: AT = alpha-tocopherol; DFE = dietary folate equivalent; mcg = microgram; mg = milligram; RAE = retinol activity equivalent (adapted from FDA fact sheet)				

RDA (MINERALS)

MINERAL	PURPOSE	FOODS		RDA
Calcium	• Blood clotting • Bone and teeth formation • Constriction and relaxation of blood vessels • Hormone secretion • Muscle contraction • Nervous system function	Almond, rice, coconut, and hemp milks Dairy products Fortified cereals	Fortified juices Fortified soy beverages Green vegetables Tofu	1,300 mg
Chloride	• Acid-base balance • Converts food into energy • Digestion • Fluid balance • Nervous system function	Celery Lettuce Olives Rye Salt substitutes	Seaweeds Table salt and sea salt Tomatoes	2,300 mg

MINERAL	PURPOSE	FOODS		RDA
Chromium	• Insulin function • Protein, carbohydrate, and fat metabolism	Broccoli Fruits Grape and orange juices Meats	Spices (garlic and basil) Turkey Whole grains	35 mcg
Copper	• Antioxidant • Bone formation • Collagen and connective tissue formation • Energy production • Iron metabolism • Nervous system function	Chocolate and cocoa Crustaceans and shellfish Lentils	Nuts and seeds Organ meats Whole grains	0.9 mg
Iodine	• Growth and development • Metabolism • Reproduction • Thyroid hormone production	Breads and cereals Dairy products Iodized salt Potatoes	Seafood Seaweed Turkey	150 mcg
Iron	• Energy production • Growth and development • Immune function • Red blood cell formation • Reproduction • Wound healing	Beans and peas Dark-green vegetables Meats Poultry Prunes and prune juice	Raisins Seafood Whole-grain, enriched, and fortified cereals and breads	18 mg
Magnesium	• Blood pressure regulation • Blood sugar regulation • Bone formation • Energy production • Hormone secretion • Immune function • Muscle contraction • Nervous system function • Normal heart rhythm • Protein formation	Avocados Bananas Beans and peas Dairy products Green leafy vegetables Nuts and pumpkin seeds	Potatoes Raisins Wheat bran Whole grains	420 mg

MINERAL	PURPOSE	FOODS		RDA
Manganese	• Carbohydrate, protein, and cholesterol metabolism • Cartilage and bone formation • Wound healing	Beans Nuts Pineapple	Spinach Sweet potato Whole grains	2.3 mg
Molybdenum	• Enzyme production	Beans and peas Nuts	Whole grains	45 mcg
Phosphorus	• Acid-base balance • Bone formation • Energy production and storage • Hormone activation	Beans and peas Dairy products Meats Nuts and seeds	Poultry Seafood Whole-grain, enriched, and fortified cereals and breads	1,2500 mg
Potassium	• Blood pressure regulation • Carbohydrate metabolism • Fluid balance • Growth and development • Heart function • Muscle contraction • Nervous system function • Protein formation	Bananas Beet greens Juices Milk Oranges and orange juice Potatoes and sweet potatoes	Prunes and prune juice Spinach Tomatoes and tomato products White beans Yogurt	4,700 mg
Selenium	• Antioxidant • Immune function • Reproduction • Thyroid function	Eggs Enriched pasta and rice Meats Nuts	Poultry Seafood Whole grains	55 mcg
Sodium	• Acid-base balance • Blood pressure regulation • Fluid balance • Muscle contraction • Nervous system function	Breads and rolls Cheese Cold cuts and cured meats Mixed meat dishes Mixed pasta dishes Pizza	Poultry Sandwiches Savory snacks Soups Table salt	2,300 mg

MINERAL	PURPOSE	FOODS		RDA
Zinc	• Growth and development • Immune function • Nervous system function • Protein formation • Reproduction • Taste and smell • Wound healing	Beans and peas Beef Dairy products Fortified cereals	Nuts Poultry Seafood Whole grains	11 mg
Note: mcg = microgram; mg = milligram (adapted from FDA fact sheet)				